N+ - Network security, maintenance, and troubleshooting procedures

THEME: STARTUP OFFICE IT SETUP

SECTION A: Scenario-Based Concept Application

1. **Explain the selection process for CPU, RAM, and storage specifications to ensure hardware compatibility and prevent performance bottlenecks for a startup team performing concurrent development and video conferencing tasks.**

Answer:

- CPU: Choose a multi-core processor (e.g., Intel Core i7 or AMD Ryzen 7) with a high clock speed. More cores = better multitasking for coding + video calls at the same time.
- RAM: Use at least 16GB DDR4 RAM. Development tools (IDEs and compilers) and video conferencing apps are memory-hungry. More RAM = smoother multitasking without slowdowns.
- Storage: Use an SSD (Solid State Drive) instead of an HDD. SSDs load applications faster. At least 256GB–512GB is recommended. Use an NVMe SSD for even better speed.
- Compatibility: Always check the motherboard's supported CPU socket type and RAM slots. Mismatched specs cause boot failures or instability.
- Tip: Match RAM speed (MHz) with the motherboard's supported speed to avoid automatic downclocking.

2. **Explain how system resource allocation and hardware bottlenecks contribute to performance degradation after the deployment of resource-intensive background applications like antivirus software.**

Answer:

- Antivirus software runs in the background and constantly scans files, which consumes CPU cycles and RAM.
- CPU Bottleneck: When both the development tasks and antivirus scans compete for CPU time, the processor gets overloaded, causing slowness.
- RAM Bottleneck: Antivirus takes up RAM. If total RAM usage exceeds physical memory, the system uses the pagefile (virtual memory on disk), which is much slower.
- Disk I/O Bottleneck: Real-time scanning reads and writes to disk constantly, slowing down other disk-dependent operations like compiling code or loading apps.
- Result: The system becomes slow, laggy, or unresponsive because hardware resources are being shared between too many processes at once.

3. **Explain the systematic troubleshooting process for resolving peripheral hardware failures caused by driver incompatibility or operating system conflicts when a device functions inconsistently across different workstations.**

Answer:

Step-by-step process:

- Step 1 – Identify the Issue: Check which workstations have the problem. Note the OS version and device model on each machine.
- Step 2 – Check Device Manager: Open Device Manager and look for yellow warning icons (!) next to the device. This shows driver errors.

- Step 3 – Update/Rollback Drivers: If recently updated, roll back the driver. If outdated, download the correct driver from the manufacturer's website.
- Step 4 – Check OS Compatibility: Confirm that the driver version supports the OS (e.g., Windows 10 vs Windows 11). Use the manufacturer's compatibility chart.
- Step 5 – Test on Another Machine: Plug the peripheral into another workstation to confirm if the issue is device-specific or workstation-specific.
- Step 6 – Reinstall Driver: Uninstall the driver completely and reinstall from scratch. Restart after each install.
- Step 7 – Check USB/Port Issues: Try a different port or cable. Some drivers conflict with specific USB controllers.

4. Explain the role of Uninterruptible Power Supply (UPS) systems and surge protection in preventing file system corruption and ensuring data integrity during unexpected power fluctuations.

<u>Answer:</u>

- UPS (Uninterruptible Power Supply): It provides battery backup power when electricity suddenly cuts off. This gives the system extra time to save data and shut down properly, preventing abrupt shutdowns.
- Surge Protector: It blocks sudden voltage spikes (surges) from reaching the computer hardware. Voltage spikes can damage HDDs, SSDs, or motherboards instantly.
- File System Corruption: When power cuts off mid-write, the file system may not finish saving data, leaving corrupted or half-written files. UPS prevents this.

- Data Integrity: With UPS, the system shuts down safely, and data is properly saved. Without it, data may be lost, or databases may become corrupted.
- Real Example: A database server losing power mid-transaction can corrupt the entire database. UPS gives it time to complete or rollback the transaction safely.

SECTION B: Practical Application

Task 1: Physically assemble a workstation and document the component installation sequence.

Answer:

Component Installation Sequence:

- Step 1 – Prepare Case: Open the computer case and ground yourself using an anti-static wristband.
- Step 2 – Install CPU: Place the CPU into the motherboard socket carefully, align the arrow marks, and lock it.
- Step 3 – Apply Thermal Paste: Apply a small pea-sized amount of thermal paste on the CPU.
- Step 4 – Install CPU Cooler: Attach the cooler on top of the CPU and secure it with screws.
- Step 5 – Install RAM: Push RAM sticks into the correct slots (check dual-channel slots in the manual).
- Step 6 – Mount Motherboard: Place the motherboard inside the case and screw it onto the standoffs.
- Step 7 – Install Storage (SSD/HDD): Connect storage to the SATA or M.2 slot on the motherboard.
- Step 8 – Install PSU (Power Supply): Mount the PSU and connect power cables to the motherboard, CPU, and storage.

- Step 9 – Install GPU (if needed): Insert the graphics card into the PCIe slot.
- Step 10 – Connect Cables: Attach front panel connectors (power button, USB, audio) to the motherboard.
- Step 11 – Power On and Test: Boot and check if the system reaches the BIOS/UEFI screen.

Task 2: Troubleshoot and resolve a RAM mismatch error within the BIOS/UEFI settings.

Answer:

- RAM mismatch happens when installed RAM runs at a different speed than what the CPU supports, or when mixing different RAM sticks.
- Step 1: Enter BIOS/UEFI – Press Del or F2 at startup.
- Step 2: Go to the Memory or DRAM settings section in BIOS.
- Step 3: Check detected RAM speed. If it shows lower than rated (e.g., 2133MHz instead of 3200MHz), enable XMP/EXPO profile.
- Step 4: Enable XMP (Extreme Memory Profile), which auto-configures RAM to its rated speed.
- Step 5: Save settings and reboot. Check if the error is resolved.
- Step 6: If still failing, reseat the RAM sticks (remove and push back firmly). Try installing one stick at a time to find a faulty module.

<u>Task 3: Build a LAN topology in Packet Tracer for 5 systems and 1 printer using a Class C subnet.</u>

<u>Answer:</u>

- Network: 192.168.1.0 / 24 (Class C Subnet)
- Device IPs:
- PC1 – 192.168.1.1 | PC2 – 192.168.1.2 | PC3 – 192.168.1.3
- PC4 – 192.168.1.4 | PC5 – 192.168.1.5 | Printer – 192.168.1.10
- Subnet Mask: 255.255.255.0 | Gateway: 192.168.1.254
- Steps in Packet Tracer:
- 1. Drag 5 PCs, 1 Printer, and 1 Switch onto the workspace.
- 2. Connect all devices to the switch using Copper Straight-Through cables.
- 3. Assign IP addresses and subnet masks to each device.
- 4. Ping each device to verify connectivity.
- Success Criteria: All pings succeed, confirming the LAN is working correctly.

<u>Task 4: Perform a clean OS installation with manual disk partitioning and startup optimization.</u>

<u>Answer:</u>

- Step 1 – Boot from USB/ISO: Set the USB drive as the first boot device in BIOS. Insert Windows/Linux ISO bootable USB.
- Step 2 – Start Installation: Select language, timezone, and keyboard layout.
- Step 3 – Choose Custom Install: Select 'Custom (Advanced)' to do manual partitioning.

- Step 4 – Manual Partitioning (Windows example):
- • Partition 1 (EFI/System): ~500MB – for boot files
- • Partition 2 (OS Drive): ~100–120GB – for Windows
- • Partition 3 (Data): Remaining space – for user files
- Step 5 – Complete Installation: Follow prompts, set username and password.
- Step 6 – Startup Optimization:
- • Open Task Manager > Startup tab > Disable unnecessary startup apps
- • Use msconfig to limit startup services
- • Keep only essential programs enabled
- Success Criteria: OS boots without errors, all partitions are accessible.

SECTION C: Integrated Case Study

Scenario: At Nexus Innovations, three rendering workstations pass morning POST checks but restart randomly during heavy afternoon workloads. The units are stored in poorly ventilated desk cabinets.

C-Q1. Identify the three hardware components most likely to cause random reboots under load.

Answer:

The three most likely culprits are:

- 1. CPU (Processor) – Overheating under heavy render workloads causes thermal throttling and auto-shutdown/restart.

- 2. GPU (Graphics Card) – Rendering tasks put an extreme load on the GPU. Overheating or power draw spikes can cause reboots.
- 3. PSU (Power Supply Unit) – High-load tasks demand more power. An underpowered or failing PSU can cause sudden restarts when it can't supply enough watts.

C-Q2. Explain why the systems pass POST in the morning but fail in the afternoon.

Answer:

- In the morning: The systems are cool (room temperature). Hardware is within safe operating temperature. POST (Power-On Self Test) completes successfully because all components are stable.
- In the afternoon: After hours of heavy rendering workload, heat builds up inside the cabinets (poor ventilation = no heat escape). CPU, GPU, and other components exceed safe temperature limits.
- Modern systems have thermal protection: When temperatures exceed a safe threshold (e.g., CPU hits 90–100°C), the system auto-restarts or shuts down to prevent permanent damage.
- This is a classic thermal throttling/emergency shutdown scenario caused by accumulated heat in enclosed spaces.

C-Q3. How would you use a stress-test tool to confirm a thermal or power issue?

Answer:

- Tools to use: Prime95 or AIDA64 (for CPU/RAM stress), FurMark (for GPU stress), HWMonitor or MSI Afterburner (for temperature monitoring).
- Step 1: Install HWMonitor to monitor real-time CPU, GPU temperatures, and voltages.
- Step 2: Run Prime95 (CPU stress test) and FurMark (GPU stress test) simultaneously to simulate heavy rendering load.
- Step 3: Watch temperature readings in HWMonitor. If the CPU crosses 90°C or the GPU crosses 85°C within minutes, it confirms a thermal issue.
- Step 4: If the system restarts during the stress test but temperatures seem okay, it may be a PSU power delivery issue — check voltage rails for instability.
- Step 5: Compare stress test results from morning (cool system) vs afternoon (heated system) to confirm heat-related pattern.

C-Q4. Propose three specific steps to remediate the frequent system restarts.

Answer:

Three Remediation Steps:

- Step 1 – Improve Ventilation: Move workstations out of enclosed desk cabinets or cut ventilation holes in the cabinets. Add external fans to ensure airflow around the units.
- Step 2 – Clean Dust & Reapply Thermal Paste: Open each workstation, clean dust from CPU heatsink, GPU fans, and all vents using compressed air. Reapply fresh thermal paste on the CPU for better heat transfer.

- Step 3 – Upgrade Cooling System: Replace stock CPU coolers with high-performance air coolers or liquid coolers (AIO). Add more case fans (intake + exhaust) to improve internal airflow.

C-Q5. Provide a brief technical justification for why a PSU upgrade might (or might not) be necessary.

Answer:

- PSU Upgrade MAY be necessary if: The current PSU wattage is insufficient for the combined load of CPU + GPU + RAM during rendering. For example, if each workstation uses a high-end GPU (e.g., RTX 4080 – 320W) plus CPU (150W) plus other components, a 600W PSU may not be enough. Upgrading to an 850W–1000W PSU provides headroom and stable power delivery.
- PSU Upgrade may NOT be necessary if: The original PSU wattage is adequate for the hardware, and the shutdowns are purely thermal (heat-related). In that case, fixing ventilation and cooling would solve the problem without a PSU replacement.
- How to check: Use a PSU wattage calculator or monitor the 12V rail voltage under load using HWMonitor. If the voltage drops below 11.4V under stress, the PSU is failing and needs upgrading.
- Conclusion: First, fix the thermal issue. Only upgrade the PSU if power instability is confirmed during stress testing.